NATIONAL UNIVERSITY
of Computer & Emerging Sciences, Karachi Campus

Assignment-III
Probability & Statistics (MT-2005)
Spring 2026
Max Marks (Wgt): 100 (04)

Ques1: **Workplace Interruptions in Tech Companies**

A recent report indicates that a typical employee in the tech industry experiences an average of 2.1 hours per day of interruptions (such as notifications, emails, and unscheduled meetings). A random sample of 50 software engineers from a large IT company showed an average interruption time of 1.8 hours per day, with a population standard deviation of 20 minutes.

Estimate the true mean interruption time with 90% confidence, and compare your result with the reported study.

Ques2: **Laptop Prices Across Online Stores**

The prices (in dollars) for a specific model of a developer laptop (with similar specifications such as RAM, processor, and storage) are listed below from 10 online retailers:

225, 240, 215, 206, 211, 210, 193, 250, 225, 202

Estimate the true mean price of this laptop model with 95% confidence.

Ques3: **Entry-Level Software Engineer Salaries**

A researcher claims that the average salary of entry-level software engineers is more than $42,000. A sample of 30 engineers shows a mean salary of $43,260. The population standard deviation is $5230.

At $\alpha = 0.05$, test the claim that entry-level software engineers earn more than $42,000 per year.

Ques4: **Revenue of Tech Companies**

A researcher estimates that the average annual revenue of large tech companies in the United States exceeds $24 billion. A sample of 50 companies is selected, and their revenues (in billions of dollars) are provided. The population standard deviation is $\sigma = 28.7$.

At $\alpha = 0.05$, use the P-value method to determine whether there is sufficient evidence to support the researcher's claim.

1	78	122	91	44	35	61	56	46	20	32	30
	28	28	20	27	29	16	16	19	15	41	38
	36	15	25								
31	30	19	19	19	24	16	15	15	19	25	25
	18	14	15	24	23	17	17	22	22	21	20
	17	20									

Ques5: **App Usage Frequency**

A report suggests that, on average, a user opens a mobile application 5.8 times per day. A researcher collects data from 20 users, as shown below:

3, 2, 1, 3, 7, 2, 9, 4, 6, 6, 8, 0, 5, 6, 4, 2, 1, 3, 4, 1

At $\alpha = 0.05$, determine whether the average app usage is still 5.8 times per day.

Ques6: **Performance of Two Algorithms**

An experiment was conducted to compare the performance degradation (wear) of two different algorithms under heavy workload conditions.

- Algorithm 1 was tested on 12 runs, yielding an average performance loss of 85 units with a standard deviation of 4.
- Algorithm 2 was tested on 10 runs, yielding an average performance loss of 81 units with a standard deviation of 5.

Assuming normal populations with equal variances, test at the 0.05 significance level whether the performance degradation of Algorithm 1 exceeds that of Algorithm 2 by more than 2 units.